FACTS: EBSI Simulator

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1. Project Overview

1.1 Project Description

Since EBSI APIs are not fully accessible for development and testing, a simulation framework has been developed in order to reproduce credential issuance, verification, and authorization flows.

This enables the development and validation of higher-level applications such as FACTS (Facts Authenticity and Credibility Tracking System).

The simulator is based on EBSI APIs documentation.

1.2 Technology Stack

Component	**Technology**
Containerization	Docker
Backend Language	Python 3
Backend Framework	FastAPI
Database	PostgreSQL with SQLModel/SQLAlchemy

1.3 EBSI Blockchain Simulation

The EBSI Blockchain is simulated using FastAPI and a PostgreSQL database, including all the CRUD actions.

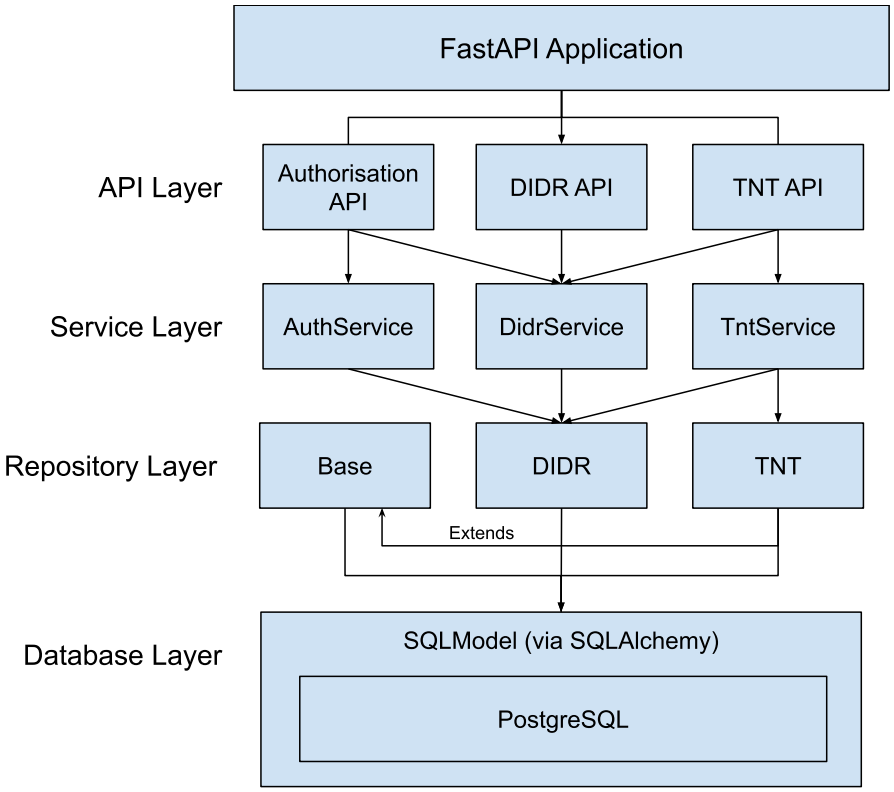
The simulation includes:

- Authorisation API (v4) - OAuth2 and OIDC implementation
- Track'n'Trace API (TNT) - Document and event tracking
- DID Registry API (v5) - Decentralized Identifier management
- Wallet Mock - Verifiable Presentation creation and ETH Transaction signature (mocking Client ETH Wallet)
- Issuer Mock - Verifiable Credential issuance (mocking an Authorized Issuer)

2. Architecture

The simulator is built following the "**Service - Repository**" design pattern.

2.1 System Architecture Diagram (excluding mock APIs)



2.2 Layer Descriptions

API Layer (ebsi_sim/api/)

- Handles HTTP requests and responses
- Implements RESTful endpoints for each service
- Handles authentication and authorization
- Input validation using Pydantic schemas

Service Layer (ebsi_sim/services/)

- Contains business logic
- Implements use cases and business rules
- Coordinates between API and Repository layers
- Handles transaction management

Repository Layer (ebsi_sim/repositories/)

- Implements Data Access Object (DAO) pattern
- Provides generic CRUD operations
- Abstracts database operations
- Returns domain models to service layer

Model Layer (ebsi_sim/models/)

- Defines database table schemas
- Uses SQLModel for ORM functionality
- Defines relationships between entities
- Includes database constraints

2.3 Project folder tree

```
📁 ebsi_sim/
├── 🐳 compose.yaml
├── 🐳 Dockerfile
├── 📄 requirements.txt
├── 📁 src/
│   ├── 🐍 __init__.py
│   ├── 🐍 main.py
│   ├── 🐍 utils.py
│   ├── 📁 api/
│   │   ├── 🐍 __init__.py
│   │   ├── 🐍 authorisation.py
│   │   ├── 🐍 didr.py
│   │   ├── 🐍 issuer_mock.py
│   │   ├── 🐍 tnt.py
│   │   └── 🐍 wallet_mock.py
│   ├── 📁 core/
│   │   ├── 🐍 __init__.py
│   │   ├── 🐍 auth.py
│   │   ├── 🐍 config.py
│   │   ├── 🐍 db.py
│   │   └── 🐍 exceptions.py
│   ├── 📁 includes/
│   │   ├── 📄 abi_didr.json
│   │   ├── 📄 abi_tnt.json
│   │   ├── 📄 presentation_didr_invite.json
│   │   ├── 📄 presentation_didr_write.json
│   │   ├── 📄 presentation_timestamp_write.json
│   │   ├── 📄 presentation_tir_invite.json
│   │   ├── 📄 presentation_tir_write.json
│   │   ├── 📄 presentation_tnt_authorise.json
│   │   ├── 📄 presentation_tnt_create.json
│   │   ├── 📄 presentation_tnt_write.json
│   │   ├── 📄 presentation_tpr_write.json
│   │   └── 📄 presentation_tsr_write.json
│   ├── 📁 models/
│   │   ├── 🐍 __init__.py
│   │   ├── 🐍 didr.py
│   │   └── 🐍 tnt.py
```

```
│   ├── 📁 repositories/
│   │   ├── 🐍 __init__.py
│   │   ├── 🐍 base.py
│   │   ├── 🐍 didr.py
│   │   └── 🐍 tnt.py
│   ├── 📁 schemas/
│   │   ├── 🐍 __init__.py
│   │   ├── 🐍 access.py
│   │   ├── 🐍 document.py
│   │   ├── 🐍 event.py
│   │   ├── 🐍 identifier.py
│   │   ├── 🐍 jsonrpc.py
│   │   ├── 🐍 presentation.py
│   │   ├── 🐍 shared.py
│   │   ├── 🐍 token.py
│   │   ├── 🐍 verifiable_credential.py
│   │   ├── 🐍 verifiable_presentation.py
│   │   └── 🐍 verification.py
│   └── 📁 services/
│       ├── 🐍 __init__.py
│       ├── 🐍 authorisation.py
│       ├── 🐍 didr.py
│       └── 🐍 tnt.py
└── 📁 tests/
    ├── 🐍 __init__.py
    ├── 🐍 conftest.py
    ├── 🐍 test_authorisation.py
    ├── 🐍 test_didr.py
    ├── 🐍 test_flow_a.py
    ├── 🐍 test_flow_b.py
    ├── 🐍 test_flow_c.py
    ├── 🐍 test_flow_d.py
    └── 🐍 test_tnt.py
```

3. API Endpoints

3.1 Authorisation API (/authorisation)

Method	Endpoint	Description
GET	/.well-known/openid-configuration	OpenID Provider Metadata
GET	/jwks	OpenID Provider's public keys
GET	/presentation-definitions	Presentation definition requirements
POST	/token	Token endpoint for OAuth2 flow

3.2 DID Registry API (/did-registry)

Method	Endpoint	Description
GET	/abi	Get smart contract ABI
GET	/identifiers	List identifiers
GET	/identifiers/{did}	Get a DID document
POST	/json-rpc	JSON-RPC endpoint

3.3 Track and Trace API (/track-and-trace)

Method	Endpoint	Description
GET	/abi	Get smart contract ABI
GET	/documents	List documents
GET	/documents/{document_id}	Get specific document
GET	/documents/{document_id}/events	List document events
GET	/documents/{document_id}/events/{event_id}	Get a document event
GET	/documents/{document_id}/accesses	List document accesses
GET	/accesses	List accesses
HEAD	/accesses	Check if DID has creator access
POST	/json-rpc	JSON-RPC endpoint

3.4 Mock APIs

Wallet Mock API (/wallet-mock)

Method	Endpoint	Description
GET	/create_vp	Create Verifiable Presentation from credential
GET	/sign_transaction	Sign Ethereum transaction

Issuer Mock API (/issuer-mock)

Method	Endpoint	Description
GET	/request_vc	Request Verifiable Credential

4. Service Layer

4.1 AuthService (ebsi_sim/services/authorisation.py)

- generate_access_token() - Generate JWT access token
- generate_id_token() - Generate JWT ID token
- load_presentation() - Load presentation definition by scope
- decode_and_check_signature() - Decode and verify token signature
- find_credential() - Find and validate credential in VP
- extract_and_validate_credentials() - Extract credentials from VP
- get_verifiable_presentation() - Decode and validate VP
- check_scope_constraints() - Validate scope constraints
- create_token() - Create access and ID tokens from VP

4.2 DidrService (ebsi_sim/services/didr.py)

- get_did_document() - Retrieve DID document by DID
- count_did_documents() - Count DID documents with optional controller filter
- list_did_documents() - List DID documents with pagination and controller filter
- get_verification_method() - Get verification method by ID
- list_verification_methods() - List verification methods with pagination
- insert_did_document() - JSON-RPC: Create new DID with verification methods and relationships
- update_base_document() - JSON-RPC: Update DID base document
- add_controller() - JSON-RPC: Add DID controller
- revoke_controller() - JSON-RPC: Revoke DID controller
- add_verification_method() - JSON-RPC: Add verification method to Identifier
- add_verification_relationship() - JSON-RPC: Add verification relationship (authentication, assertionMethod, etc.)
- revoke_verification_method() - JSON-RPC: Revoke verification method (set expiration)
- expire_verification_method() - JSON-RPC: Set expiration date for verification method
- get_abi() - Get DID Registry contract ABI
- handle_rpc() - Handle JSON-RPC requests with scope and access validation

JSON-RPC Scope Requirements:

Method	Required Scopes
insertDidDocument	didr_invite, didr_write
updateBaseDocument	didr_write
addService	didr_write
revokeService	didr_write
addController	didr_write
revokeController	didr_write
addVerificationMethod	didr_write
addVerificationRelationship	didr_write
revokeVerificationMethod	didr_write
expireVerificationMethod	didr_write
rollVerificationMethod	didr_write
sendSignedTransaction	didr_invite, didr_write

4.3 TntService (ebsi_sim/services/tnt.py)

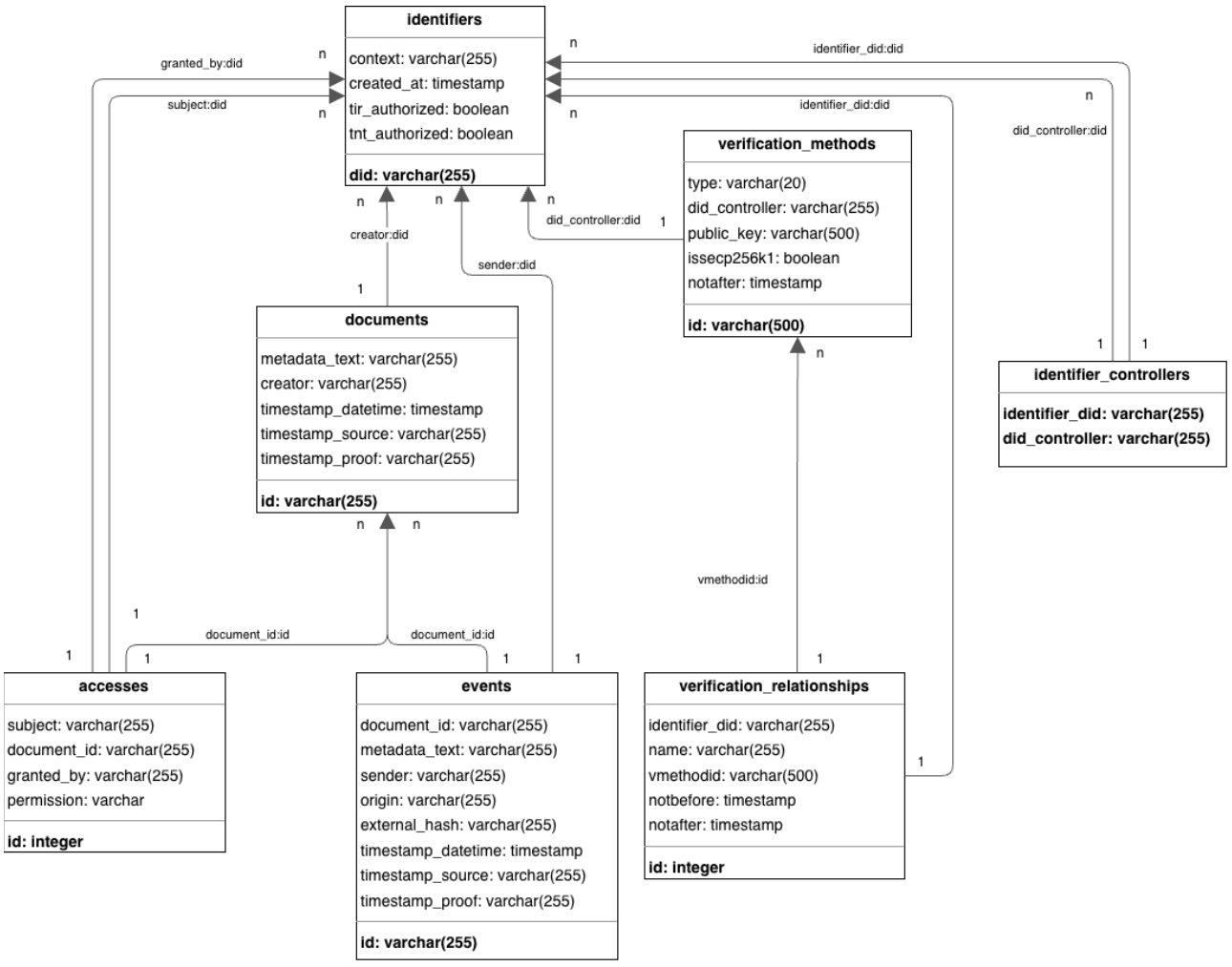
- get_document() - Retrieve document by hash
- count_documents() - Count documents with filters
- list_documents() - List documents with pagination
- count_accesses() - Count accesses with filters
- list_accesses() - List accesses with pagination
- count_events() - Count events with filters
- list_events() - List events with pagination
- authorise_did() - JSON-RPC: Authorize/unauthorize DID for TNT operations
- create_document() - JSON-RPC: Create new document with metadata and timestamps
- remove_document() - JSON-RPC: Remove document (creator only)
- grant_access() - JSON-RPC: Grant write/delegate access to document
- revoke_access() - JSON-RPC: Revoke document access
- write_event() - JSON-RPC: Write event to document with access control
- get_abi() - Get TNT contract ABI
- handle_rpc() - Handle JSON-RPC requests with scope and method constraint validation

JSON-RPC Scope Requirements:

Method	**Required Scopes**
authoriseDid	tnt_authorise
createDocument	tnt_create
removeDocument	tnt_write
grantAccess	tnt_write
revokeAccess	tnt_write
writeEvent	tnt_write
sendSignedTransaction	tnt_authorise, tnt_create, tnt_write

5. Database structure

5.1 Relational model



6. Exception Handling

6.1 Core Exceptions (ebsi_sim/core/exceptions.py)

- **EBSIError**: Base exception for all EBSI errors (Status Code: 500)
- **EBSIDatabaseError**: Database-related errors (Status Code: 500)
- **EBSIDuplicateError**: Duplicate entry errors (Status Code: 409)
- **EBSIAuthError**: Authentication errors (Status Code: 401)
- **EBSIRequestError**: Request processing errors (Status Code: 400)
- **EBSINotFoundError**: Resource not found errors (Status Code: 404)

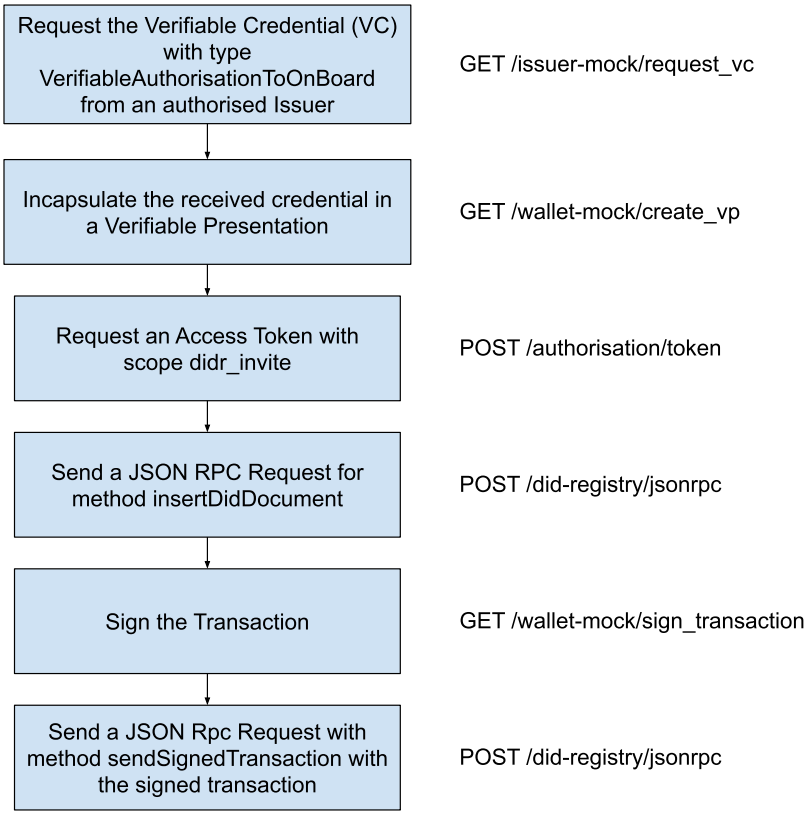
6.2 Service-Specific Exceptions

- **AuthServiceError**: Base error for Auth service operations (Status Code: 500)
- **AuthServiceAuthError**: Authentication error in Auth service (Status Code: 401)
- **AuthServiceNotFoundError**: Resource not found in Auth service (Status Code: 404)
- **AuthServiceRequestError**: Request error in Auth service (Status Code: 400)
- **DidrServiceError**: Base error for DID Registry service (Status Code: 500)
- **DidrServiceDuplicateError**: Duplicate entry in DID Registry service (Status Code: 409)
- **DidrServiceAuthError**: Authentication error in DID Registry service (Status Code: 401)
- **DidrServiceNotFoundError**: Resource not found in DID Registry service (Status Code: 404)
- **DidrServiceRequestError**: Request error in DID Registry service (Status Code: 400)
- **TntServiceError**: Base error for TNT service (Status Code: 500)
- **TntServiceDuplicateError**: Duplicate entry in TNT service (Status Code: 409)
- **TntServiceAuthError**: Authentication error in TNT service (Status Code: 401)
- **TntServiceNotFoundError**: Resource not found in TNT service (Status Code: 404)
- **TntServiceRequestError**: Request error in TNT service (Status Code: 400)

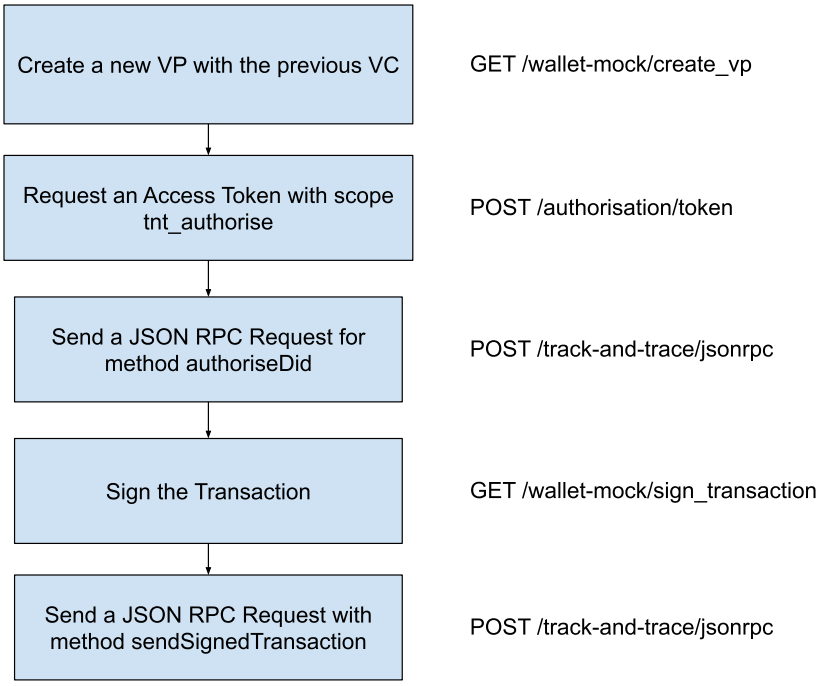
7. Simulator Base Flows

7.1 Flow A: Create DID Document

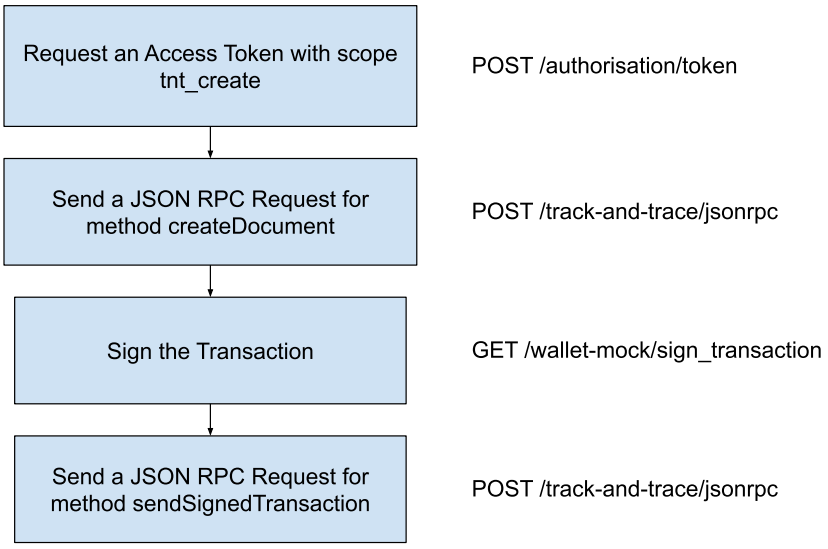
Most important flow, it is the onboarding of a new DID in the DID Registry



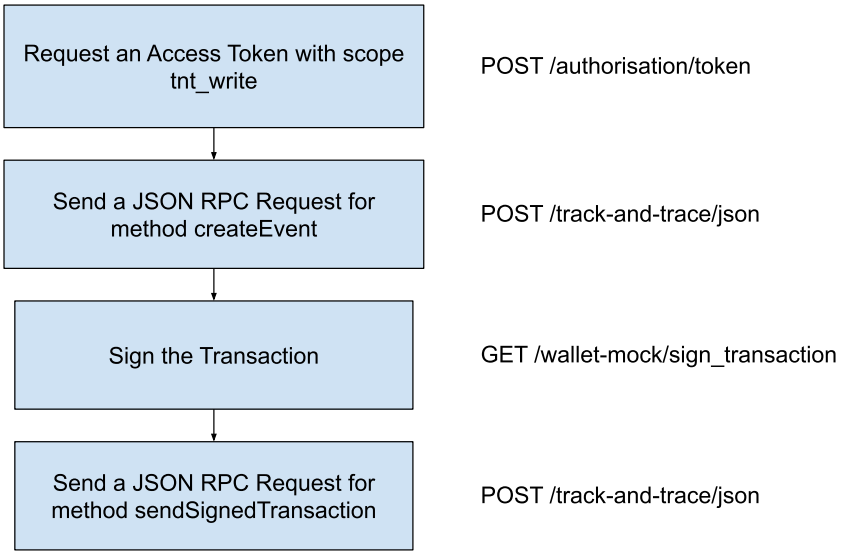
7.2 Flow B: Get authorization to operate on TNT



7.3 Flow C: Create TNT Document



7.4 Flow D: Add event to a TNT Document



8. Testing

8.1 Test Files

Test File	Description
conftest.py	Pytest fixtures configuration (client and mock db)
test_authorisation.py	Tests for Authorisation API endpoints
test_didr.py	Tests for DID Registry API endpoints
test_tnt.py	Tests for Track'n'Trace API endpoints
test_flow_a.py	Tests flow A CRUD endpoints
test_flow_b.py	Tests flow B CRUD endpoints
test_flow_c.py	Tests flow C CRUD endpoints
test_flow_d.py	Tests flow D CRUD endpoints

8.2 Test Coverage

The test suite covers:

- API endpoint response status codes
- OIDC configuration discovery
- Ethereum ABI retrieval
- Presentation definitions retrieval
- TNT document retrieval and listing
- TNT document events retrieval and listing
- TNT document accesses retrieval and listing
- DID document retrieval and listing
- Authentication endpoints
- CRUD Operations for the different user flows

8.3 Tests

test_authorisation.py

Test	Description
test_read_oidc_config	Tests the retrieval of the OpenID Connect configuration via the designated endpoint.
test_read_presentation_definitions_tir_write	Tests the retrieval of presentation definitions specific to the `tir_write` scope using the provided test client.
test_read_presentation_definitions_tnt_write	Tests the retrieval of presentation definitions specific to the `tnt_write` scope using the provided test client.

test_read_presentation_definitions_tpr_write	Tests the retrieval of presentation definitions specific to the `tpr_write` scope using the provided test client.
test_read_presentation_definitions_didr_invite	Tests the retrieval of presentation definitions specific to the `didr_invite` scope using the provided test client.
test_read_presentation_definitions_didr_write	Tests the retrieval of presentation definitions specific to the `didr_write` scope using the provided test client.
test_read_presentation_definitions_timestamp_write	Tests the retrieval of presentation definitions specific to the `timestamp_write` scope using the provided test client.
test_read_presentation_definitions_tir_invite	Tests the retrieval of presentation definitions specific to the `tir_invite` scope using the provided test client.
test_read_presentation_definitions_tnt_authorise	Tests the retrieval of presentation definitions specific to the `tnt_authorise` scope using the provided test client.
test_read_presentation_definitions_tnt_create	Tests the retrieval of presentation definitions specific to the `tnt_create` scope using the provided test client.
test_read_presentation_definitions_tsr_write	Tests the retrieval of presentation definitions specific to the `tsr_write` scope using the provided test client.
test_read_presentation_definitions_fake	Tests the retrieval of presentation definitions with a non-existing scope using the provided test client.

test_didr.py

Test	**Description**
test_read_abi	Tests the retrieval of the DID Registry ABI.
test_read_identifiers	Tests the retrieval of all identifiers.
test_read_identifier	Tests the retrieval of a specific identifier.
test_read_identifier_not_found	Tests the retrieval of a non-existent identifier.

test_tnt.apy

Test	Description
test_read_abi	Tests the retrieval of the Track and Trace ABI.
test_head_accesses	Tests the check of having access to TNT API with an existent DID
test_head_accesses_not_found	Tests the check of having access to TNT API with a non-existent DID
test_read_accesses	Tests the retrieval of accesses for a specific DID
test_read_documents	Tests the retrieval of all documents.
test_read_document	Tests the retrieval of a specific document.
test_read_document_not_found	Tests the retrieval of a non-existent document.
test_read_document_events	Tests the retrieval of events for a specific document.
test_read_document_not_found_events	Tests the retrieval of events for a non-existent document.
test_read_document_event	Tests the retrieval of a specific event for a document.
test_read_document_event_not_found	Tests the retrieval of a non-existent event for a document.
test_read_document_accesses	Tests the retrieval of accesses for a specific document.
test_read_document_not_found_accesses	Tests the retrieval of accesses for a non-existent document.

test_flow_a.py

Test	Description
test_request_vc	Tests the request_vc endpoint of the Issuer Mock API
test_create_vp	Tests the create_vp endpoint of the Wallet Mock API
test_access_token_didr_invite	Tests the access token request for the didr_invite scope on Authorisation API
test_insert_did_document_create_transaction	Tests the creation of a transaction to insert a DID document on DIDR API
test_insert_did_document_sign_transaction	Tests the signing of a transaction on Wallet Mock API

test_insert_did_document_send_signed_transaction	Tests the sending of an insertDidDocument signed transaction to DIDR API

test_flow_b.py

Test	**Description**
test_create_vp_with_vmethod	Tests the creation of a verifiable presentation with a VerificationMethod on Wallet Mock API
test_access_token_tnt_authorise	Tests the access token request for the tnt_authorise scope on Authorisation API
test_tnt_authorise_did_create_transaction	Tests the creation of a transaction to authorise a DID to TNT API
test_tnt_authorise_send_signed_transaction	Tests the sending of an authoriseDid signed transaction to TNT API

test_flow_c.py

Test	**Description**
test_access_token_tnt_create	Tests the access token request for the tnt_create scope on Authorisation API
test_tnt_document_create_transaction	Tests the creation of a transaction to create a TNT Document on TNT API
test_tnt_document_send_signed_transaction	Tests the sending of an createDocument signed transaction to TNT API

test_flow_d.py

Test	Description
test_access_token_tnt_write	Tests the access token request for the tnt_write scope on Authorisation API
test_tnt_write_event_create_transaction	Tests the creation of a transaction to create a TNT Event on TNT API
test_tnt_write_event_send_signed_transaction	Tests the sending of a writeEvent signed transaction to TNT API

9. Configuration

9.1 Environment Variables (ebsi_sim/core/config.py)

Setting	Description
DB_USER	PostgreSQL username
DB_PASSWORD	PostgreSQL password
DB_HOST	PostgreSQL host
DB_PORT	PostgreSQL port
DB_NAME	PostgreSQL database name
AUTH_PUBLIC_KEY	Authorization service public key (ES256)
AUTH_PRIVATE_KEY	Authorization service private key (ES256)
ETH_ADDRESS	Ethereum address
ISSUER_AUTHENTICATION_PUBLIC_KEY	Issuer authentication public key (ES256K)
ISSUER_AUTHENTICATION_PRIVATE_KEY	Issuer authentication private key (ES256K)
ISSUER_ASSERTION_PUBLIC_KEY	Issuer assertion public key (ES256)
ISSUER_ASSERTION_PRIVATE_KEY	Issuer assertion private key (ES256)
ISSUER_DID	Issuer DID

9.2 Docker configuration (ebsi_sim/core/config.py)

The project includes Docker configuration for easy deployment and development:

- Dockerfile - Containers definition for the EBSI Simulator
- compose.yaml - Docker Compose configuration for multi-container setup